Math 10C — Unit 5: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

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1. Complete the following statements.

(a) The first coordinate of an ordered pair is called the _____ coordinate, and the second coordinate is called the _____ coordinate.

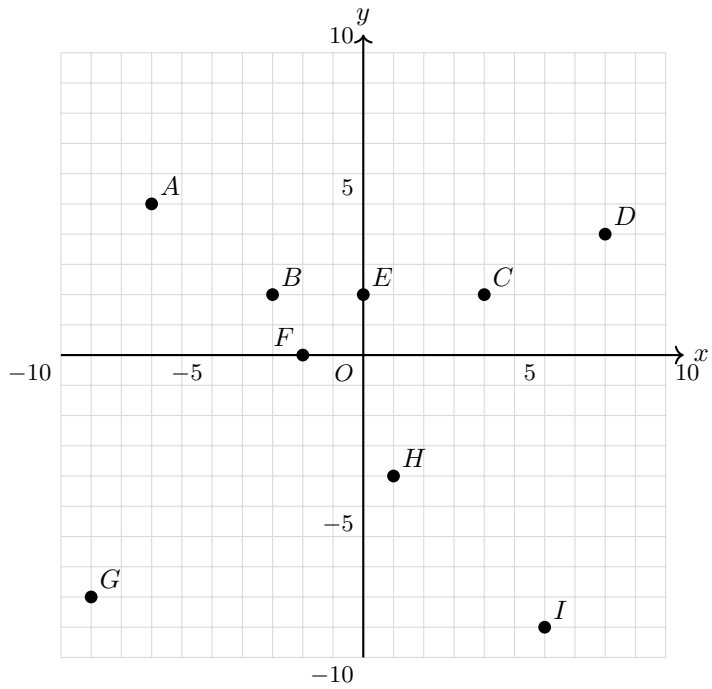
(b) _____ _____ are used to locate or plot a point on a Cartesian plane.

(c) The two numbers in an ordered pair are together called the _____ of a point on the grid.

(d) The x-axis and the y-axis intersect at the _____.

(e) The Cartesian plane is divided into four _____.

2. The following questions refer to the grid below.



(a) Name the point represented by each of the following coordinates.
i) $(-7, 5)$ ii) $(6, -9)$ iii) $(0, 2)$

(b) List the coordinates of each point.
i) C ii) G iii) F iv) O v) H

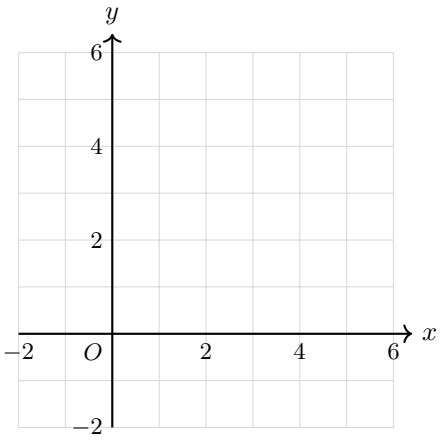
(c) Which points lie in
i) quadrant I ii) quadrant II iii) quadrant III iv) quadrant IV

(d) Which points lie in between quadrants?

(e) Which points have the same x-coordinate? What visual check can be used?

(f) Which points have the same y-coordinate? What visual check can be used?

3. Plot the following points, in order, on the grid below and join each point to the next with a straight line segment: $(0, 0)$ to $(4, 0)$ to $(4, 3)$ to $(2, 5)$ to $(0, 3)$ and back to $(0, 0)$. What geometric figure do the points outline?



4. Use the following information to answer this question.

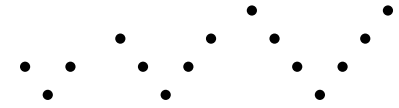


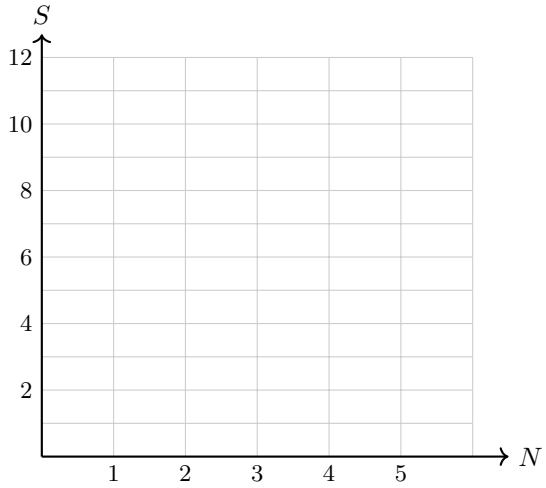
Figure 1 Figure 2 Figure 3

A V-shaped pattern of dots is shown. Figure 1 has 3 dots: one at the bottom of the V and one on each arm. Each new figure adds one more dot to the end of each arm.

(a) Complete the table relating the number of dots, S, to the figure number, N.

Figure Number, N	1	2	3	4	5
Number of Dots, S					

(b) Represent the data from the table on the grid.



(c) Does it make sense to join the points with a straight line? Explain.

(d) Describe in words the relationship between the number of dots and the figure number.

(e) Write an equation that can be used to determine the number of dots, if we know the figure number.

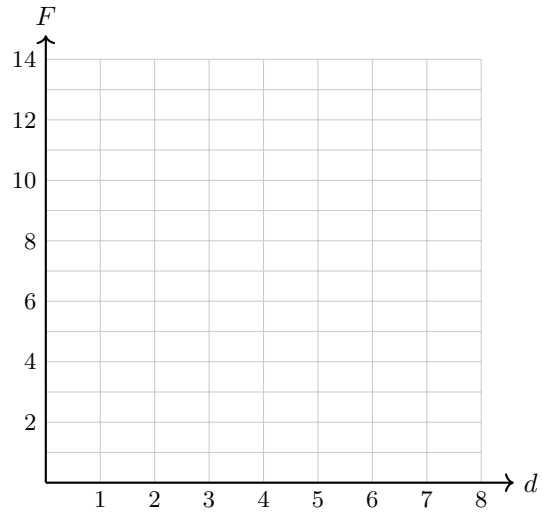
(f) Use the equation to determine the number of dots in figure 50.

(g) Use the equation to determine the figure number, if there are 137 dots.

5. A taxi company charges a fixed pickup fee plus an additional amount for each kilometre travelled. The table below shows the fare, F dollars, for a trip of d kilometres.

Distance, d **(km)**	0	2	4	6	8
Fare, F **(\$)**	3.50	5.90	8.30	10.70	13.10

(a) Represent the data on the grid.



(b) Does it make sense to join the points together with a straight line? Explain.

(c) Use the graph to estimate the fare for a trip of 5 km.

(d) Use the graph to estimate the distance travelled if the fare was $11.90.

(e) Write an equation expressing F in terms of d.

(f) Use the equation to check the accuracy of your estimates in (c) and (d).

(g) In the context of the question, what do the values 3.50 and 1.20 represent in your equation?

6. A staircase pattern is built from 1 cm by 1 cm squares. Each new diagram is one step higher and one square wider than the one before.

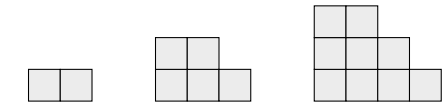


Diagram 1 Diagram 2 Diagram 3

(a) Find the perimeter of Diagrams 1–3 by counting edges, then extend the pattern to complete the table relating the perimeter, P, to the diagram number, D.

Diagram Number, D	1	2	3	4	5
Perimeter (cm), P					

(b) Does it make sense to join the points with a straight line? Explain.

(c) Write an equation that can be used to determine the perimeter, if we know the diagram number.

(d) Use the equation to determine the perimeter of diagram 10.

(e) Use the equation to determine the diagram number, if the perimeter is 78 cm.

7. Describe, in general, how to decide whether the points on the graph of a relation should or should not be connected.

8. *(Multiple Choice)* Which of the following variables is discrete?

A. weight of a truck
B. temperature outside
C. number of pets owned
D. distance travelled

9. *(Multiple Choice)* Which of the following variables is continuous?

A. number of siblings
B. height of a plant
C. number of correct answers on a quiz
D. number of books on a shelf

10. *(Multiple Choice)* Consider the following variables: i) the diameter of a tree trunk ii) the number of rings in a tree trunk iii) the height of a tree iv) the number of trees in a forest
Which of the variables above is/are continuous?

A. i) only
B. i) and iii) only
C. i), ii), and iii) only
D. all of i), ii), iii), and iv)

11. *(Multiple Choice)* On a Cartesian plane, the line segment joining the points $(-4, -3)$ and $(6, -6)$

A. intersects both the x-axis and y-axis.
B. intersects the y-axis but not the x-axis.
C. intersects the x-axis but not the y-axis.
D. does not intersect the x-axis or the y-axis.

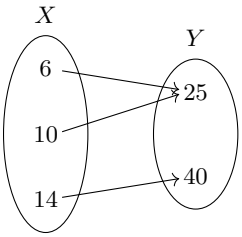
12. *(Numerical Response)* The relationship between degrees Celsius, C, and degrees Fahrenheit, F, is described by the equation $C = \frac{5}{9}(F - 32)$. Normal human body temperature is 98.6°F.
To the nearest degree, normal human body temperature in degrees Celsius is _____.

Record your answer left to right. | | | | |

Lesson 2 Assignment *Show your work.*

1–3. For each relation below, state **(i)** the independent variable, **(ii)** the dependent variable, **(iii)** the list of inputs, and **(iv)** the list of outputs.

1.



2.

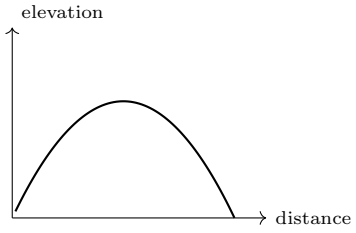
p	q
4	11
9	26
14	41

3. (a, b):
$(3, 14),\ (6, 26),\ (9, 38),\ (12, 50)$

4. For each relation below, state **(i)** the independent variable and **(ii)** the dependent variable.

(a) A worker's wages, w (dollars), are related to the number of hours worked, h, by the equation $w = 6.5h$.

(b)



(c) The number of bacteria, n, in a culture is related to the time, t (in hours), since the culture began growing.

5. Consider the linear relation described by the equation $y = 2x - 4$.

(a) Determine the value of y for each input in the table.

x	-3	-1	0	2	5
y					

(b) Determine the value of x that produces an output of $y = 10$.

(c) Is this relation linear or nonlinear? Explain how you can tell from the equation.

6. Consider the nonlinear relation described by the equation $y = x^2 - 5$.

(a) Determine the value of y for each input in the table.

x	-3	-1	0	2	4
y					

(b) Given that the output is 11 when $x = 4$, use the symmetry of the graph to predict the output when $x = -4$.

(c) Verify your answer to part (b) by substituting $x = -4$ into the equation.

(d) Is this relation linear or nonlinear? Explain how you can tell from the equation.

7. In each table the inputs go up in equal steps. Find the change in y from each output to the next, then state whether the relation is **linear** or **nonlinear**.

(a)

x	0	1	2	3	4
y	5	8	11	14	17

(b)

x	0	1	2	3	4
y	1	2	5	10	17

(c)

x	0	2	4	6	8
y	10	7	4	1	-2

8. A candle is 20 cm tall when lit and burns down at a steady rate of 1.5 cm per hour.

(a) Write an equation for the height, h (cm), of the candle after t hours of burning.

(b) Identify the independent variable and the dependent variable.

(c) Determine the height of the candle after 6 hours.

(d) Determine how many hours it takes for the candle to burn down to a height of 5 cm.

9. A gym charges a one-time \$40 sign-up fee plus \$15 per month of membership.

(a) Determine the total cost, C, for $m = 1, 2, 3$, and 4 months of membership.

m	1	2	3	4
C				

(b) Write an equation relating the total cost C to the number of months m.

(c) Identify the independent variable and the dependent variable, and explain your choice.

10. *(Multiple Choice)* "The temperature of a cup of coffee, T, is related to the time, t, since it was poured." The independent variable in this relation is

A. the temperature of the coffee
B. the time since it was poured
C. the rate of cooling
D. the room temperature

11. *(Multiple Choice)* Which ordered pair belongs to the relation $y = 4x - 9$? A. $(2, -1)$ B. $(3, 2)$ C. $(0, 9)$ D. $(4, 5)$

12. *(Numerical Response)* The table below shows a linear relation. Determine the value of y when $x = 20$.

x	2	5	8	11
y	9	18	27	36

Record your answer left to right. ☐☐☐☐

13. *(Numerical Response)* For the relation $y = 5x - 2$, let A be the output when the input is 7, and let B be the input that produces an output of 48. The value of $A + B$ is _____.

Record your answer left to right. ☐☐☐☐

Lesson 3 Assignment *Show your work.*

1. Determine the **value** of the y-intercept of the graph of each equation.

(a) $y = x - 8$ (b) $y = 4x - 20$ (c) $3y + 4x - 12 = 0$

(d) $0.4x - 1.5y + 3 = 0$ (e) $3y = x^2 - 75$ (f) $y = 0.002x^2 - 0.002x + 8.36$

2. Determine the **value** of the x-intercept(s) of the graph of each equation.

(a) $y = x - 6$ (b) $y = 3x - 9$ (c) $2y + 5x - 20 = 0$

(d) $0.8x - 4y + 1.6 = 0$ (e) $y = x^2 - 16$ (f) $y = 20 - 4x$

3. Determine the x- and y-intercepts of each equation. **Write each intercept point as an ordered pair.**

(a) $y = 5x + 20$ (b) $y = 24 - 8x$ (c) $6x - 3y + 18 = 0$

(d) $y = \dfrac{x^2}{3} - 12$ (e) $y = 9 - x^2$ (f) $y = 5x$

(g) $y = x^2 + 9$ (h) $y = x^2 - 4x$ (i) $y = -2x^2 + 8$

4. Northern Lights Moving Company charges a flat fee of \$150 plus \$0.25 per km for a long-distance move. The total cost, C dollars, of a move of n km is $C = 150 + 0.25n$.

(a) Complete the table of values.

n (km)	0	800	2000	3200
C (\$)				

(b) What does the ordered pair $(0, 150)$ represent?

(c) Determine the n-intercept of the graph algebraically. Explain why it is not applicable in the context of this problem.

(d) Use the formula to determine the cost of a move of

(i) 1200 km (ii) 5000 km

(e) If a move costs a total of \$800, determine the distance travelled.

5. A water tank initially holds 2400 L. It drains at a constant rate of 60 L per minute. The volume, V litres, remaining after t minutes is $V = 2400 - 60t$.

(a) Complete the table of values.

t (min)	0	10	20	30	40
V (L)					

(b) How much water is in the tank before draining begins?

(c) How many minutes does it take for the tank to drain completely?

(d) How much water remains in the tank after 25 minutes?

(e) How long will it take for the tank to have 300 L remaining?

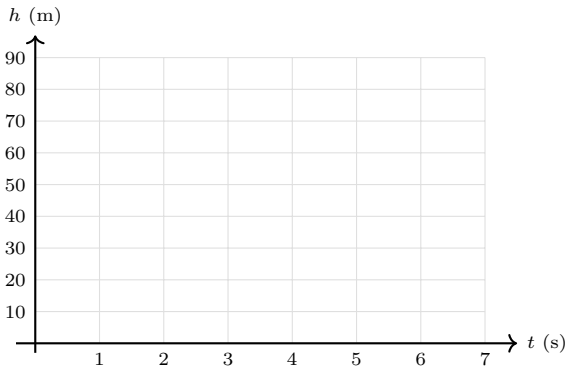
(f) Is this relation linear or nonlinear? Explain how you know.

6. A ball is launched upward from a platform. Its height, h metres, above the ground after t seconds ($t \geq 0$) is modelled by $h = -5t^2 + 30t + 35$.

(a) The maximum height is reached after 3 seconds. Calculate the maximum height.

(b) Complete the table of values, then sketch the graph of h versus t on the grid, joining the points with a smooth curve.

t (s)	0	1	2	3	4	5	6	7
h (m)								



(c) Is this relation linear or nonlinear?

(d) For how many seconds is the ball in the air?

(e) What does the h-intercept represent in the context of this problem?

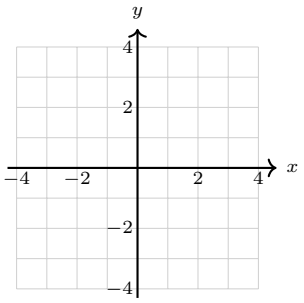
(f) What does the t-intercept represent in the context of this problem?

(g) Use the equation to calculate the exact height of the ball after 4.5 seconds.

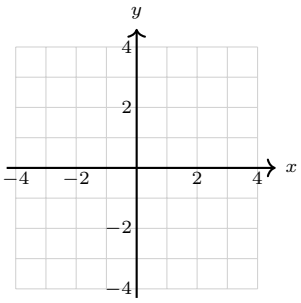
(h) Does it make sense to extend the graph further in a downward direction to the left or to the right? Explain.

7. Sketch a line with the given intercepts on each grid. For (b), (c) and (d), also write the equation of the line.

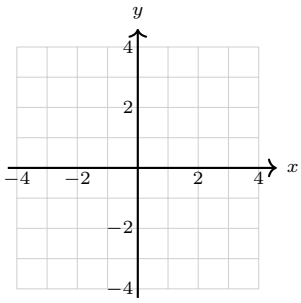
(a) x-intercept 4 and y-intercept -2



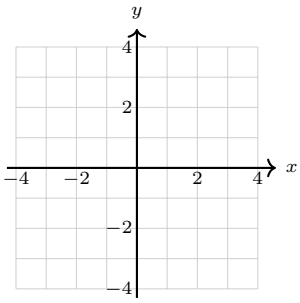
(b) y-intercept -3 and no x-intercept



(c) x-intercept 3 and no y-intercept



(d) infinitely many y-intercepts



8. *(Multiple Choice)* In which of the following relations does the graph have x- and y-intercepts with equal values?

A. $y = x + 6$ B. $3x + 3y = 12$ C. $3x - 4y + 8 = 0$ D. none of the above

9. *(Numerical Response)* The line $3x - 4y + 24 = 0$ has x-intercept a and y-intercept b. The value of $b - a$ is _____.

Record your answer left to right. | | | | |

Extension — intercepts of other curves

10. *(Extension)* Determine the x- and y-intercepts of each equation. Write each intercept point as an ordered pair.

(a) $4x^2 + y^2 = 64$

(b) $4x^2 - y^2 = 64$

(c) $x^2 + y^2 = 49$

11. *(Extension)* The graph of $25x^2 + 16y^2 - 400 = 0$ has x-intercepts a and b, and y-intercepts c and d. Determine the value of the product $abcd$.

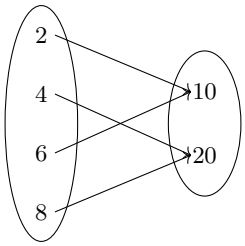
Lesson 4 Assignment *Show your work.*

1. List the domain and range of each set of ordered pairs.
(a) $(1,4),\ (3,4),\ (5,8),\ (7,-2),\ (-1,4)$ (b) $(-6,2),\ (-6,5),\ (-6,9),\ (-6,-3)$

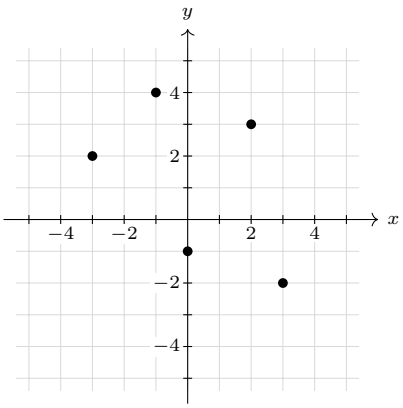
2. State the domain and range of the relation given by the table of values below.

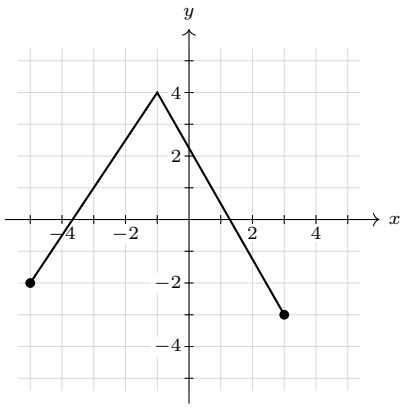
x	-3	-1	0	2	4	6
y	10	6	4	0	-4	-8

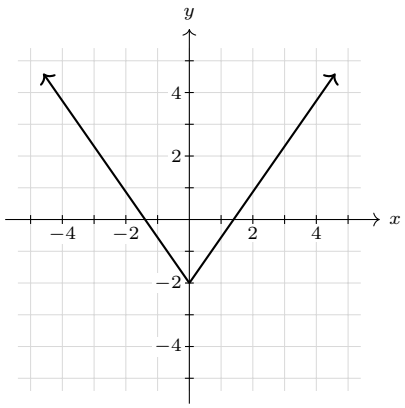
3. State the domain and range of the relation shown in the arrow diagram below.



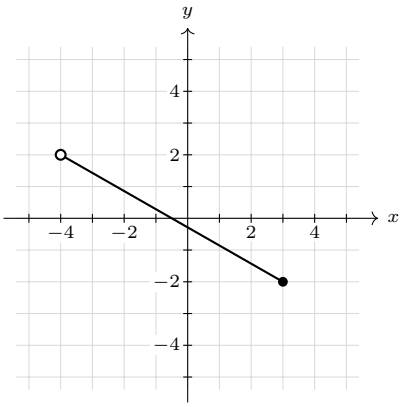
4. State the domain and range of each relation graphed below.
(a) (b) (c)



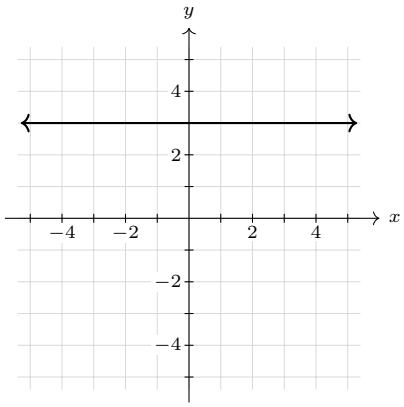




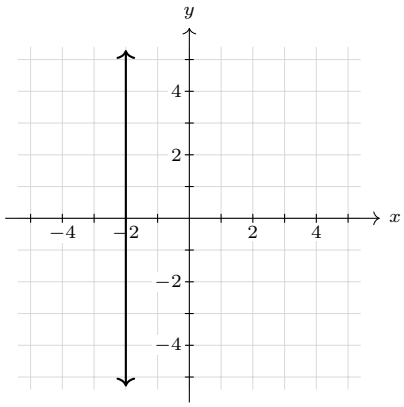
(d)



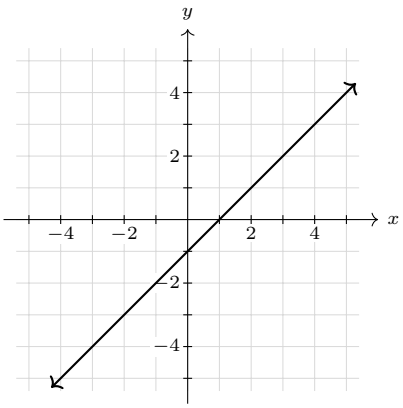
(e)



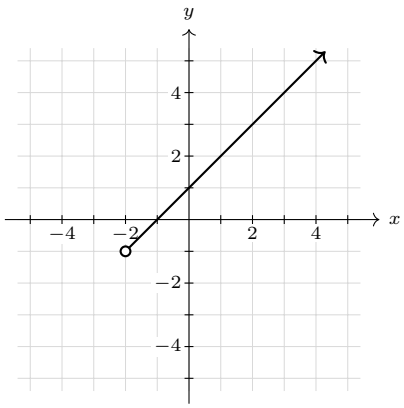
(f)



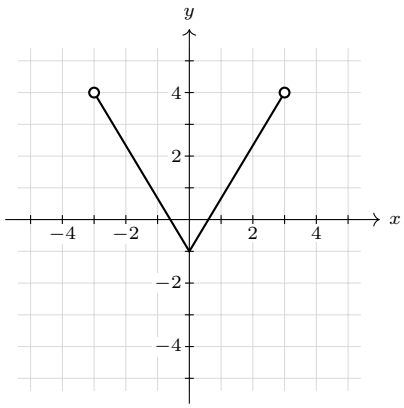
(g)



(h)

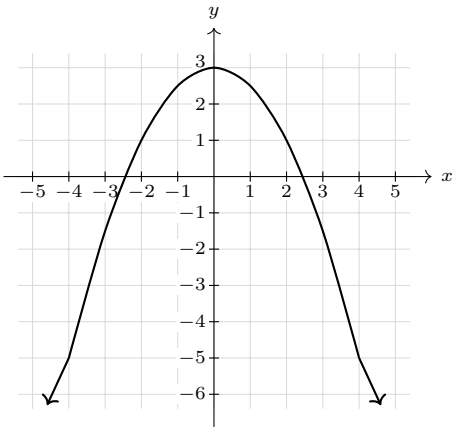


(i)

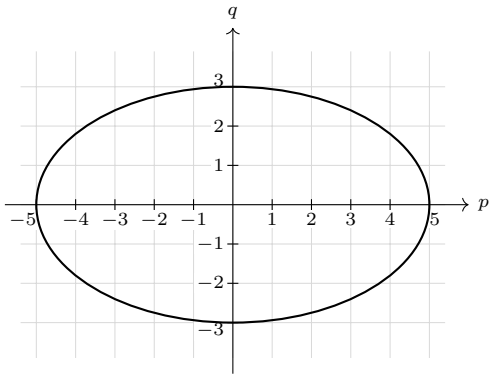


5. State the domain and range of each relation described below.

(a)



(b) An ellipse, with horizontal axis p and vertical axis q.



(c) A relation that assigns each of the values $-2,\ 0$, and 3 to the single output value 6.

(d) A circle with centre $(-4, 1)$ and a radius of 5.

6. For each relation, state whether the domain and range are best described as a finite list of values or with an inequality, then write the domain and range using the appropriate notation.

(a) $\{(-2, 3),\ (0, 3),\ (2, 3),\ (4, 3)\}$

(b) All points on the line $y = 2x - 1$ for $-3 \leq x \leq 5$.

(c) All points on the graph of $y = x^2$.

7. A courier company charges a flat fee of \$8 plus \$3 per kilometre travelled, so the cost C, in dollars, of a delivery that travels d kilometres is $C = 8 + 3d$. To keep routes efficient the company only accepts deliveries with $2 \leq d \leq 20$ kilometres.

(a) Complete a table of values for $d = 2,\ 6,\ 10,\ 15,\ 20$.

d	2	6	10	15	20
C					

(b) State the domain and range of the relation, and explain why each is restricted.

8. *(Multiple Choice)* Which of the following relations has range $\{y \mid y \leq 4,\ y \in \mathbb{R}\}$?

A. A circle centred at the origin with radius 4.

B. A downward-opening curve with maximum point $(0, 4)$ that continues infinitely far down in both directions.

C. The set of points $\{(1, 4),\ (2, 4),\ (3, 4)\}$.

D. A horizontal line segment from $(-4, 0)$ to $(4, 0)$.

9. *(Numerical Response)* A relation includes the ordered pairs $(-5, 12),\ (-2, 7),\ (0, 7),\ (3, 12),\ (6, 20)$. If a is the number of elements in the domain and b is the number of elements in the range, the value of $a \times b$ is _____.

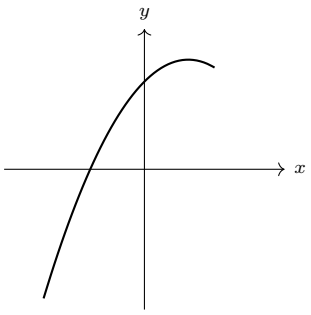
Record your answer left to right. | | | | |

Lesson 5 Assignment *Show your work.*

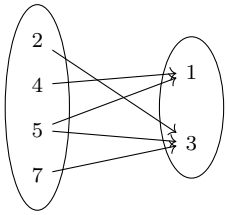
1. Determine which of the following relations are functions. Give a reason for each answer.

(a) $(-3, 2),\ (-1, 5),\ (4, 2),\ (6, 8)$

(b)



(c)



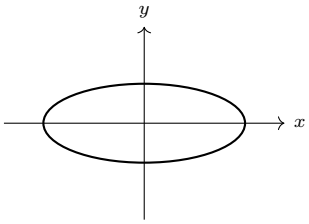
(d)

Input (x)	Output (y)
3	4
0	6
-2	7
3	9

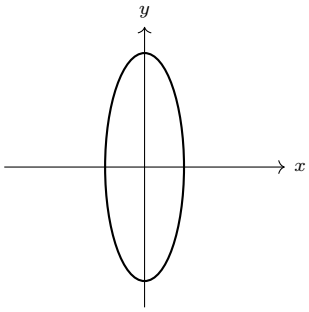
2. State which of the following relations are functions.

(a) $(0, 0),\ (2, 3),\ (4, 5),\ (6, 7),\ (8, 5)$

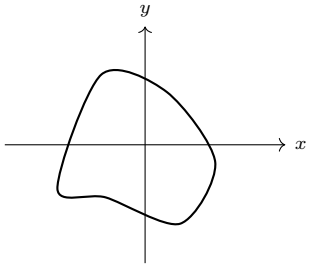
(b)



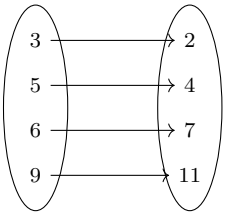
(c)



(d)



(e)

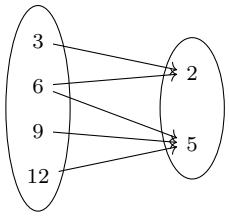


3. State which of the following relations are functions.

(a)

x	y
1	9
3	9
5	9
7	9

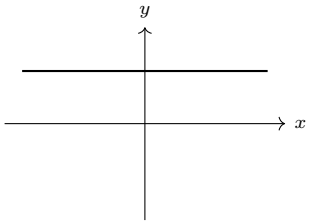
(b)



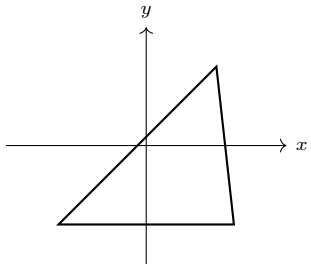
(c)

x	y
2	10
−2	10
4	10
−4	10

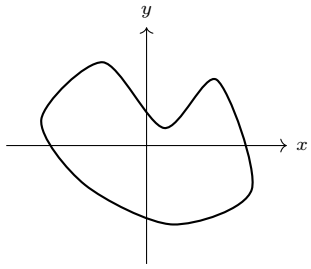
(d)



(e)



(f)



4. Mrs. X has two sons, Aiden and Blake. Mrs. Y has three children: Chloe, Danielle and Evan.

(a) Draw an arrow diagram to illustrate the relation "is the mother of" from the set of mothers to the set of children. Is this relation a function?

(b) Draw an arrow diagram to illustrate the relation "is the child of" from the set of children to the set of mothers. Is this relation a function?

5. *(Multiple Choice)* Which of the following statements is **not** always true for a function?

A. A function is a set of ordered pairs (x, y) in which every x-value corresponds to exactly one y-value.

B. Every function passes the vertical line test.

C. For every output value there is only one input value.

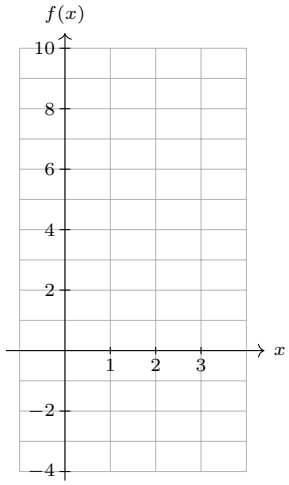
D. Every element of the domain is related to exactly one element of the range.

6. Explain why every function is a relation, but not every relation is a function. Use one relation from Question 3 that is a function and one that is not as your examples.

Lesson 6 Assignment *Show your work.*

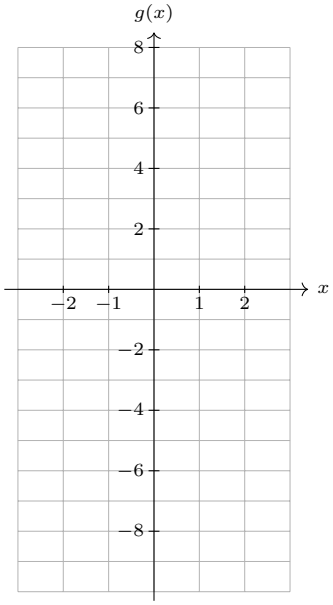
1. The function $f(x) = 4x - 3$ has domain $\{0, 1, 2, 3\}$.
(a) List the elements of the range of the function.

(b) Graph the function f.



2. The function $g(x) = x^3 - 1$ has domain $\{-2, -1, 0, 1, 2\}$.
(a) List the elements of the range of the function.

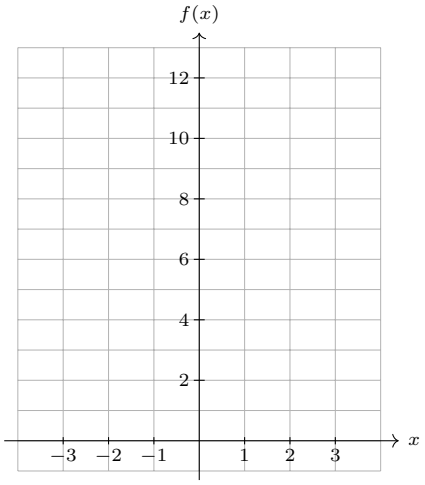
(b) Graph the function g.



3. Consider the function $f(x) = x^2 + 3$.
(a) Complete a table of values for domain $\{3, 2, 1, -1, -2, -3\}$.

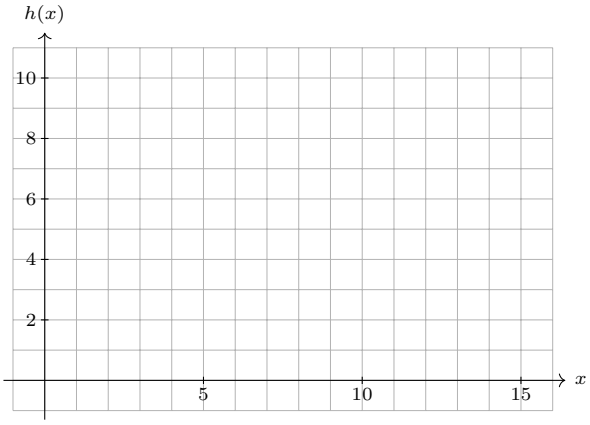
x	3	2	1	-1	-2	-3
$f(x)$						

(b) Plot the ordered pairs on the grid.
(c) Draw a smooth curve through the points to illustrate the function $f(x) = x^2 + 3$, $x \in \mathbb{R}$.



4. The function $h(x) = 9$ has domain $\{0, 5, 15\}$.
(a) List the ordered pairs of the graph of the function.

(b) Graph the function h.



5. Each statement below refers to a function h whose graph has equation $y = h(x)$. Circle the correct choice in each case.
(a) h is the *name / value* of the function.
(b) The values of x represent the *inputs / outputs* of the function.
(c) The values of $h(x)$ represent the *inputs / outputs* of the function.

(d) The values of y represent the *inputs / outputs* of the function.
(e) x represents the *independent / dependent* variable of the function.
(f) $h(x)$ represents the *independent / dependent* variable of the function.
(g) y represents the *independent / dependent* variable of the function.

6. If $f(x) = 4x - 9$, determine:
(a) $f(3)$
(b) $f(-2)$
(c) $f(0)$

7. Function g is defined by $g(x) = 7 - x^2$. Evaluate:
(a) $g(3)$
(b) $g(-5)$
(c) $g(\sqrt{2})$

8. A function f is defined by the formula $f(x) = x^3 - 2$. Find:
(a) the image of 3 under f (b) the value of f at -4

9. If $f(x) = x^3 + 3x^2 - x + 4$, evaluate:
(a) $f(2)$ (b) $f(-3)$

10. Consider the function f defined by $f(x) = 12 - 3x$, $x \in \mathbb{R}$. Determine:
(a) $f(3)$ (b) the value of f when $x = -5$ (c) the image of 1.5 under f

(d) an expression for $f(3t)$ (e) an expression for $f(a - 2)$

11. Solve for the unknown in each case.
(a) If $f(x) = 4x - 9$, determine x if $f(x) = 19$. (b) If $g(x) = 8x + 5$, determine x if $g(x) = -27$. (c) If $g(t) = 48 - 5t$, determine t if $g(t) = 13$.

(d) If $h(x) = -4x + 6$, determine x if $h(x) = 30$. (e) If $P(x) = 60 - 4x^2$, determine the values of x if $P(x) = -40$.

12. Consider the function f defined by $f(x) = 9x - 20$. Find:
(a) $f(0)$ (b) an expression for $f(3x - 1)$ (c) the solution to the equation $f(x) = 43$

13. A function C is defined by $C(x) = \sqrt{x}$, where $x \geq 0$. Evaluate (a)(i)–(iii), then answer (b).
(a)(i) $C(25)$ (a)(ii) $C\left(\frac{1}{49}\right)$ (a)(iii) $\frac{C(144)}{C(9)}$ (b) If $C(x) = 11$, find x.

14. A function g is defined by the formula $g(t) = t + 20$.
Calculate the value of $g(6) + g(-3)$.

15. A taxi fare, in dollars, for a trip of d kilometres is given by the linear function $C(d) = 3.50 + 1.20d$.
(a) Evaluate $C(12)$. (b) Explain what $C(12)$ means in this situation. (c) Solve $C(d) = 15.50$. (d) What does $C(0) = 3.50$ represent?

16. *(Multiple Choice)* The function $f(x) = 10 - 3x$ has domain $\{0, 1, 2, 3, 4\}$. Which of the following is **not** an element of the range of the function?

A. -2 B. 4 C. 7 D. 5

17. *(Multiple Choice)* If $f(x) = 2x - 1$ and $f(t) = 15$, then $t =$

A. 7 B. 8 C. 16 D. 29

18. *(Numerical Response)* A function f is defined by the formula $f(x) = 6\sqrt{x}$, $x \geq 0$. The value of $f(45)$ can be written in the form $k\sqrt{5}$. The value of k is _____.

Record your answer left to right. | | | | |

Extension — substituting expressions into non-linear functions

19. *(Extension)* For $f(x) = x^3 - 2$ from Question 8, find an expression for $f(a)$.

20. *(Extension)* For $g(t) = t + 20$ from Question 14, if $g(a^2) = 68$, determine the values of a in simplest radical form.

21. *(Extension)* If $F(x) = 2x^2 - 5x - 3$, determine a simplified expression for:

(a) $F(-x)$ (b) $F(x + 4)$

Lesson 7 Assignment *Show your work.*

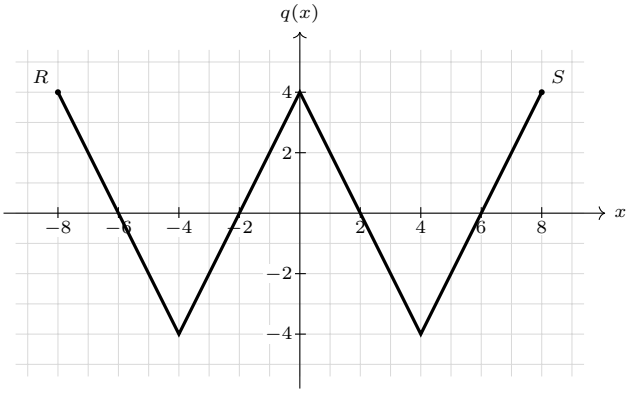
1. In each case express the relation given in function notation as an equation in two variables.
(a) $f(x) = 12 - 5x$ (b) $g(x) = 3x^2 + 7$ (c) $Q(t) = 4t - 15$

2. Express the following relations in function notation.
(a) $y = 6x + 13$ (b) $y = 8w - 21$ (c) $3x + 4y - 12 = 0$

3. Express each equation in function notation.
(a) The graph of the function defined by $y = f(x)$ has equation $y = 0.25x + 1.75$.

(b) The graph of the distance function defined by $d = f(t)$ has equation $d = 55t$ (in kilometres, for travel at a constant speed).

4. The graph of a function q is shown below.



(a) Complete:
i) $q(-6) =$ ____
ii) $q(-1) =$ ____
iii) $q(5) =$ ____

(b) Write the ordered pairs associated with i), ii) and iii).

(c) State the value(s) of x if **i)** $q(x) = 4$, **ii)** $q(x) = -4$, **iii)** $q(x) = 6$.

(d) Use the notation in (a) to make a statement about the points R and S on the graph.

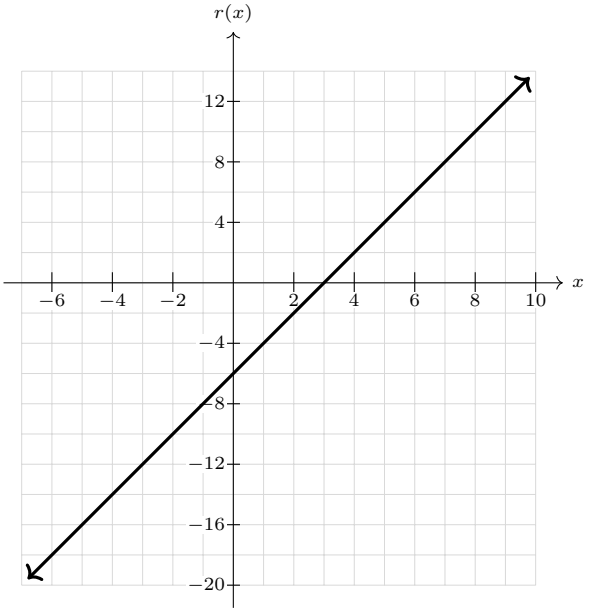
(e) Write the x- and y-intercepts of the graph using function notation.

(f) Complete:
the domain of q is
$\{x \mid$ ____ $\leq x \leq$ ____, $x \in \mathbb{R}\}$;
the range of q is
$\{q(x) \mid$ ____ $\leq q(x) \leq$ ____, $q(x) \in \mathbb{R}\}$.

5. The function $g(x) = 2x^2 - 7$ has domain $\{-2, -1, 0, 1, 2\}$.
(a) State the range of g.

(b) Solve the equation $g(x) = 1$.

6. Consider the graph of the linear function r shown below.



(a) Complete the table.

x	$r(x)$	Ordered Pair
9		
	0	
−6		
6		
	−12	
0		

(b) Use the graph to solve $r(x) = 10$. Explain why this equation has exactly one solution, unlike the zigzag graphs in Question 4.

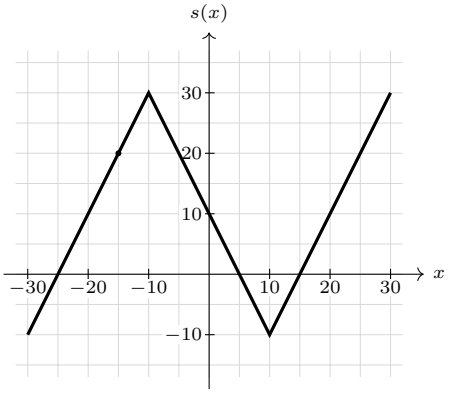
7. Given that $f(x) = 11 - 3x$,

(a) evaluate $f(-4)$,

(b) find the value of $f(t) + f(-t)$,

(c) calculate the x-intercept and the y-intercept on the graph of f.

8. The graph of a function s is shown below. *Each grid line represents* 5 *units on both axes.*



(a) A student is asked to make a statement about the marked point on the graph. The student states that $t(-3) = 4$.

i) Explain **two** errors in the student's statement.

ii) Write a correct statement using function notation about this point.

(b) Give the solution to the following equations.

i) $s(x) = -10$ **ii)** $s(x) = 30$

(c) State the value of **i)** $s(-30)$, **ii)** $s(30)$.

(d) State the domain and range of the function.

(e) The equation $s(a) = b$ has **exactly two** solutions. Explain clearly how to use the graph to determine values of a and b, and provide two sets of answers to the problem.

9. *(Extension)* Consider the function $k(x) = 2 - x^2$. Given that $k\left(a^{1/4}\right) = -23$, calculate the value of a.

10. *(Multiple Choice)* The graph of the function $w(x) = 5x + 1$, $x \in \mathbb{R}$, intersects the y-axis at
A. $(0, 0)$ B. $(0, 1)$ C. $(0, 5)$ D. $\left(-\frac{1}{5}, 0\right)$

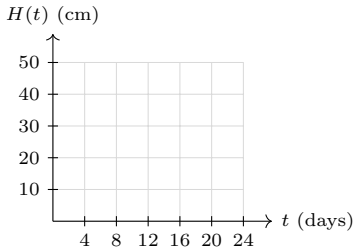
11. *(Multiple Choice)* Function T is such that $T(8) = -3$. Devon states: "When the domain value is 8, the related range value is -3." Priya states: "The point $(-3, 8)$ is on the graph of $y = T(x)$." Which of the following is true?
A. Both statements are correct.
B. Both statements are incorrect.
C. Devon is correct and Priya is incorrect.
D. Priya is correct and Devon is incorrect.

12. *(Numerical Response)* Consider the graph of the function $k(x) = 8x - 20$. The x-intercept of the graph of k is located at $(a, 0)$. The value of a is _____.

Record your answer left to right. | | | | |

13. A greenhouse seedling is treated with a special nutrient solution. The seedling's height is modelled by $H(t) = \frac{7}{4}t + 5$, where $H(t)$ is the height of the seedling, in cm, t days after treatment begins.

(a) Sketch the graph of the function on the grid provided.



(b) Determine the value of $H(8)$.

(c) Write in words the meaning of $H(8)$.

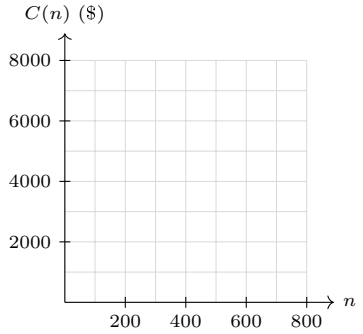
(d) Evaluate the following.
i) $H(0)$ **ii)** $H(16)$ **iii)** $H(24)$

(e) How many days will it take for the seedling to reach a height of 26 cm?

(f) The seedling reaches maturity 24 days after treatment begins. State the domain and range of the function $H(t)$.

14. Fresh Print Co. produces custom t-shirts for local sports teams. The cost of production is modelled by $C(n) = 2150+6.75n$, where $C(n)$ is the cost, in dollars, of printing n t-shirts.

(a) Sketch the graph of the function for a maximum of 800 t-shirts.



(b) Determine the value of $C(60)$.

(c) Write in words the meaning of $C(60)$.

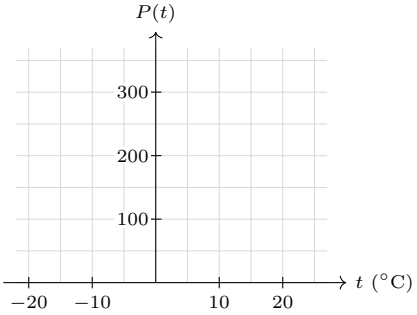
(d) Evaluate $C(0)$ and explain its significance.

(e) How many t-shirts can be printed for \$6200?

(f) Last month the company printed 500 t-shirts. **In addition to the costs in** $C(n)$, the company paid \$850 for design work and \$1200 in other monthly costs. If each t-shirt sells for \$18.50, how much profit would be made if all the t-shirts are sold?

15. A ski resort tracks how the number of hot cocoa packets sold in its lodge depends on the average daily temperature. The relationship is modelled by $P(t) = 180 - 8t$, where $P(t)$ is the number of packets sold when the average daily temperature is $t\,^\circ$C.

(a) Sketch the graph of the function on the grid provided.



(b) Determine the value of $P(-10)$.

(c) Write in words the meaning of $P(-10)$.

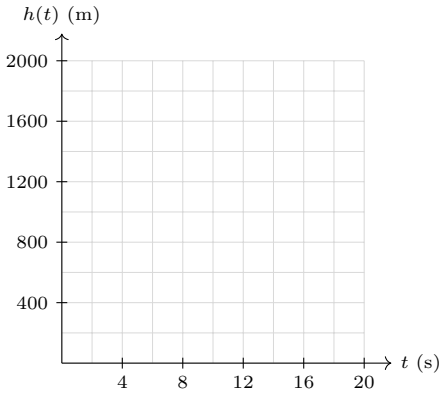
(d) What was the average temperature if 140 packets were sold?

(e) Explain how you could estimate the lower limit of the domain of this relation.

(f) Suggest an appropriate domain and range for the function $P(t)$ if the resort is located in the Canadian Rockies.

16. As part of a design test, a 12 kg equipment case is dropped from a hovering helicopter. The height of the case above the ground is described by part of the graph of $h(t) = 1960 - 4.9t^2$, where $h(t)$ is the height, in metres, of the case t seconds after it is released.

(a) Sketch the graph of the function on the grid provided.



(b) Determine the value of $h(15)$.

(c) Why does $h(25)$ not represent the height of the case 25 seconds after release?

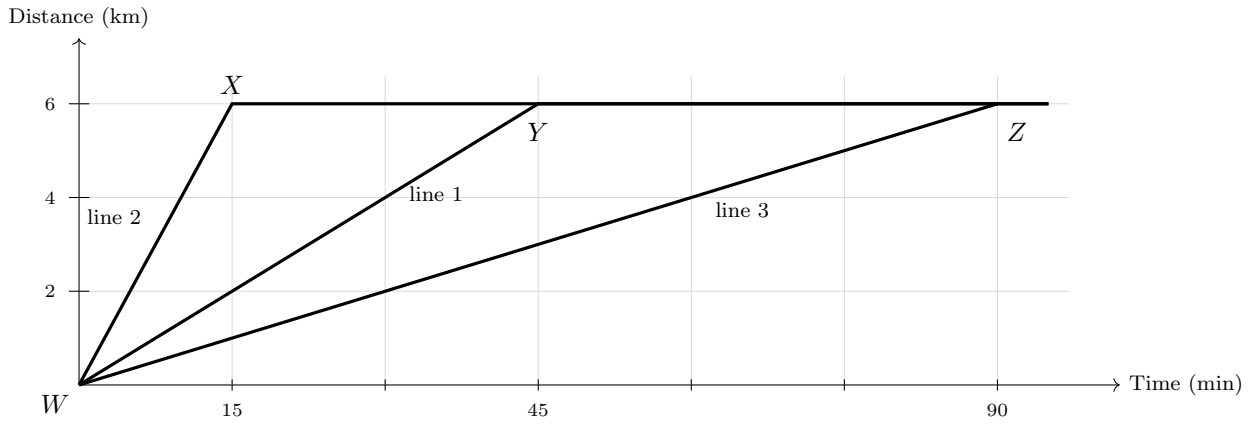
(d) At what height above the ground was the case released?

(e) How long (to the nearest second) will it take the case to reach the ground?

(f) Suggest an appropriate domain and range for the function $h(t)$.

Lesson 8 Assignment *Show your work.*

1. Three friends — Derek, Farah and Grace — each travel the same 6 km route from the mall to the stadium at a constant speed. Derek cycles, Farah jogs and Grace walks, and all three leave the mall at the same time. Lines 1, 2 and 3 represent the three trips.



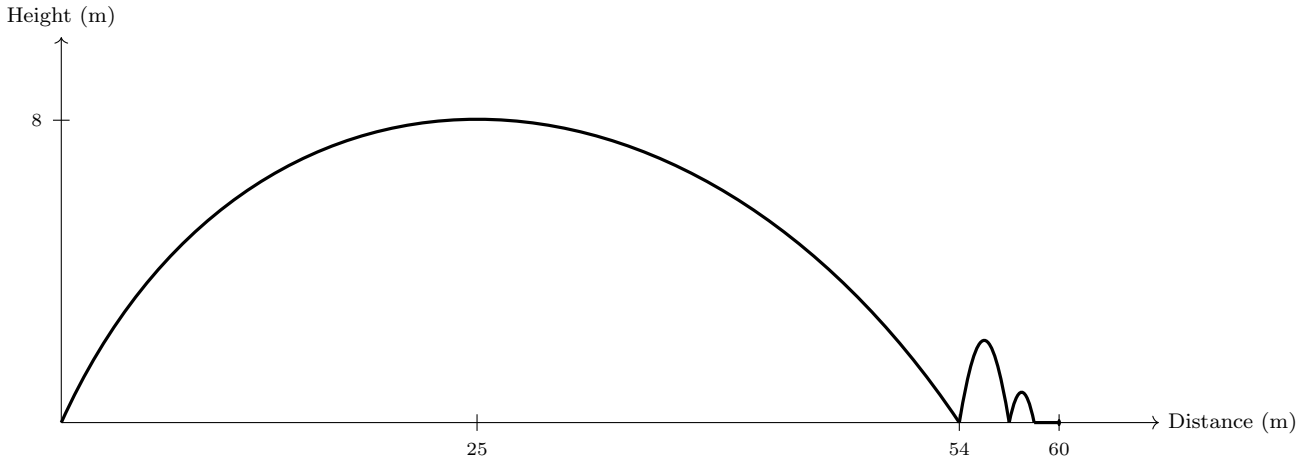
(a) Complete the table.

	Line 1	Line 2	Line 3
Distance (km)	6	6	6
Time (hrs)	$\frac{3}{4}$		
Rate (km/hr)	8		
Student			

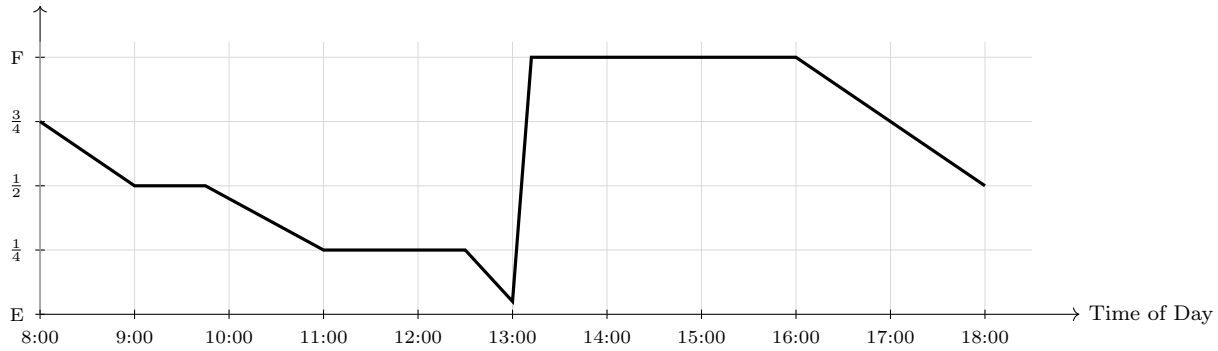
(b) Explain what is happening at each of the following points.
(i) W (ii) X (iii) Y (iv) Z

(c) How can you tell, just from the steepness of the lines, which line belongs to which person?

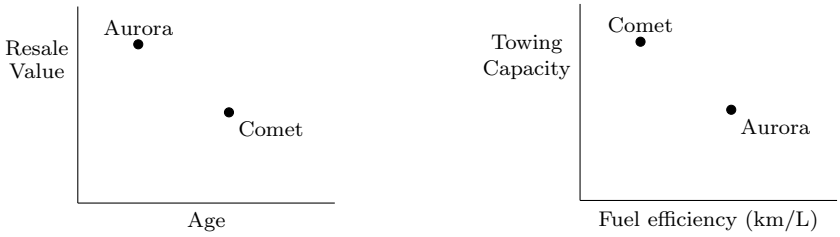
2. Priya kicks a soccer ball. The graph shows the height of the ball against the horizontal distance it travels before rolling to a stop. Describe Priya's kick, using the graph.



3. Lin delivers flowers for a shop. The graph shows the amount of gasoline in her delivery van on a particular day. Describe how Lin may have spent her day.



4. Two hybrid SUVs, the Aurora and the Comet, are compared on the two graphs below: the first by age and resale value, the second by towing capacity and fuel efficiency (in km/L, so a larger value means the vehicle goes farther on each litre). Describe the comparison between the two vehicles.

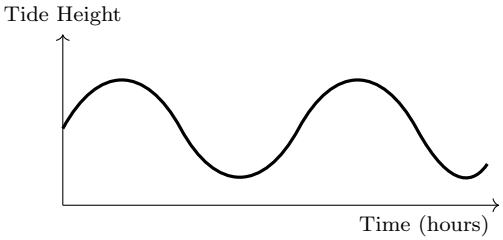


5. A rubber ball is dropped from a height of 12 m. On each bounce it rebounds to 65% of its previous height. Sketch a graph of the ball's height as a function of time, showing at least the first four bounces.

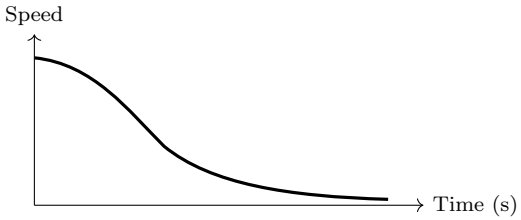


6. Suggest a possible real-world scenario for each graph below.

(a)



(b)

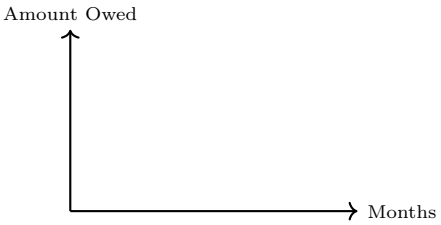


7. Sketch a graph with no scale for each of the following, on the axes provided.

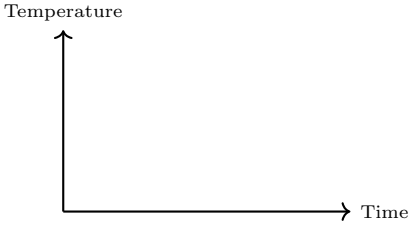
(a) The resale value of a used mountain bike as a function of its age.



(b) The amount still owed on a $480 loan, if $60 is paid off every month, as a function of the number of months, until it is fully paid off.



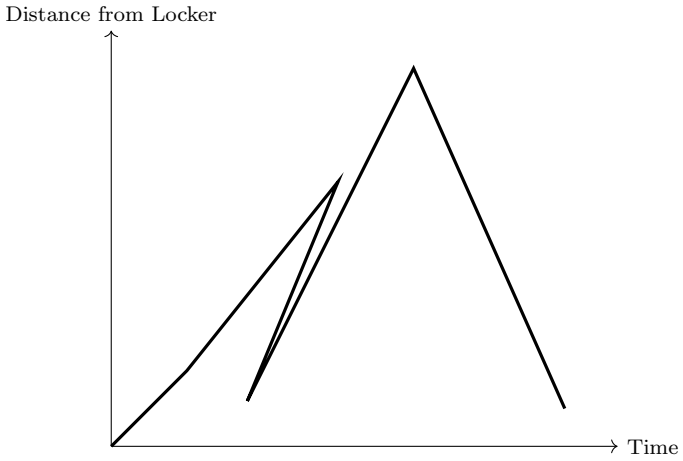
(c) The outdoor temperature on a winter day, from 6:00 a.m. to 6:00 p.m.



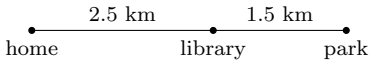
(d) Your height above the ground while riding a swinging pirate-ship carnival ride, during three full swings.



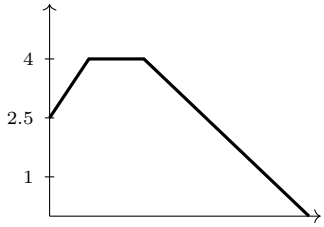
8. A student sketched the graph below to represent her distance from her locker during part of a school day. Explain why this graph cannot be correct.



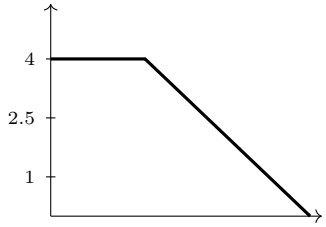
9. *(Multiple Choice)* Jordan leaves the library and cycles to the park, 1.5 km away. He spends some time at the park, then jogs straight home. The library is 2.5 km from Jordan's home, along the same route. Which graph best describes Jordan's distance from home, starting from when he left the library?



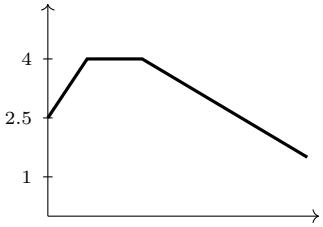
A.



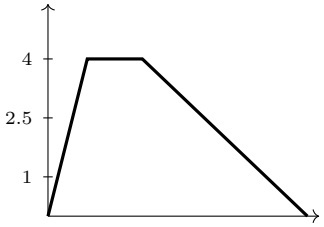
B.



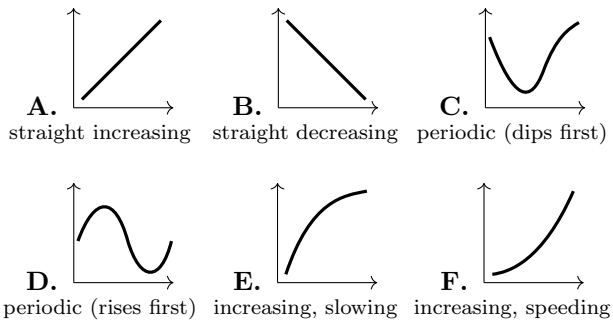
C.



D.



Use the following graphs to answer Questions 10–13. Match each description with the best graph. Each graph may be used once, more than once, or not at all.



10. A tree's height as a function of its age.

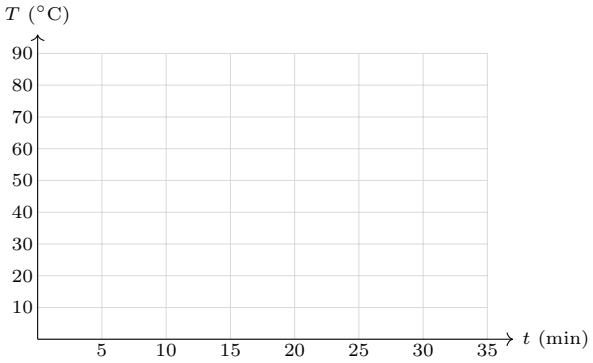
11. The average monthly rainfall in a coastal town is lowest in mid-summer and highest in winter. Rainfall as a function of month, starting in January.

12. The volume of a cube as a function of its side length.

13. You fill a bathtub at a constant rate with the drain closed. The water depth as a function of time.

14. A cup of hot chocolate is left on a counter, and its temperature is recorded every 5 minutes.

Time, t (min)	0	5	10	15	20	25	30
Temperature, T (°C)	80	62	49	40	34	30	27



(a) Graph the data on the grid.
(b) Should the points be joined? Explain.

(c) State a realistic domain and range for the data, and explain why the graph should not be extended far to the right.

15. *(Multiple Choice)* Which situation is best represented by a graph that is **constant**, then **decreasing at a constant rate**?

A. the distance from home of a parked car that then drives away at a steady speed
B. the height of a candle that sits unlit for an hour, then is lit and burns steadily
C. the height of a ball thrown straight up
D. the balance of a savings account that earns interest

16. *(Numerical Response)* In Question 1, Grace arrives at the stadium _____ minutes after Derek.

Record your answer left to right. | | | | |

Lesson 9 Assignment *Show your work.*

1. *(Multiple Choice)* Consider the function $h(x) = 7x - 21$. The x-intercept of the graph of h is
   A. -21 B. -3 C. 3 D. 21

2. *(Multiple Choice)* Given a function h defined by $h(x) = mx + n$, with $h(0) = -3$ and $h(2) = 5$, then
   A. $m = 4,\ n = -3$ B. $m = -3,\ n = 4$ C. $m = 1,\ n = -3$ D. $m = 4,\ n = 0$

3. *(Multiple Choice)* If $f(x) = 2^x$, then $f(-5) =$
   A. -32 B. -10 C. $\frac{1}{32}$ D. $\frac{1}{10}$

4. *(Multiple Choice)* If $g(x) = \frac{3}{4}x + 8$, an expression for $g(3x - 2)$ is
   A. $\frac{9}{4}x + 6$ B. $\frac{9}{4}x + \frac{13}{2}$ C. $\frac{9}{4}x + 8$ D. $\frac{9}{4}x - \frac{3}{2}$

5. *(Multiple Choice)* Consider the following four functions.
   1. $p(x) = x^2 + 3x - 1$ **2.** $p(x) = \frac{1}{2}x + 9$ **3.** $p(x) = 2x^2 - x$ **4.** $p(x) = 5 - 4x$
   Evaluate $p(-2)$ for each function and put the four results in order from greatest to least. The order is
   A. 4321 B. 4312 C. 3421 D. none of the above

6. *(Multiple Choice)* "The cost of a pizza, C, depends on the number of toppings, t." The independent variable is
   A. the cost, C B. the number of toppings, t
   C. the price of one topping D. the size of the pizza

7. *(Multiple Choice)* Which of the following representations of a relation might **fail** to describe a function — that is, it cannot guarantee that each input produces exactly one output?
   A. A table of values listing each input exactly once.
   B. A set of ordered pairs in which the first coordinate is repeated, paired with two different second coordinates.
   C. An arrow diagram in which each input has exactly one arrow leaving it.
   D. An equation solved explicitly for y in terms of x.

8. *(Numerical Response)* For the function $f(x) = 4x - 7$, $f(a) = 25$. The value of a is _____.

Record your answer left to right. | | | | |

9. *(Numerical Response)* The line $y = -3x + 12$ has x-intercept a and y-intercept b. The value of $a + b$ is _____.

Record your answer left to right. | | | | |

10. During an air show, the flight path of a stunt dive by a small plane can be modelled by $h = t^2 - 8t + 84$, where h is the height in metres after t seconds. The dive ends when the plane returns to its starting height. Complete the table of values below, then use it to answer (a) to (e).

t (s)	0	1	2	3	4	5	6	7	8
h (m)									

(a) What is the plane's height at the start of the dive?

(b) What is the lowest height the plane reaches during the dive?

(c) At what time does the plane reach this lowest point?

(d) What is the plane's height 2 seconds into the dive?

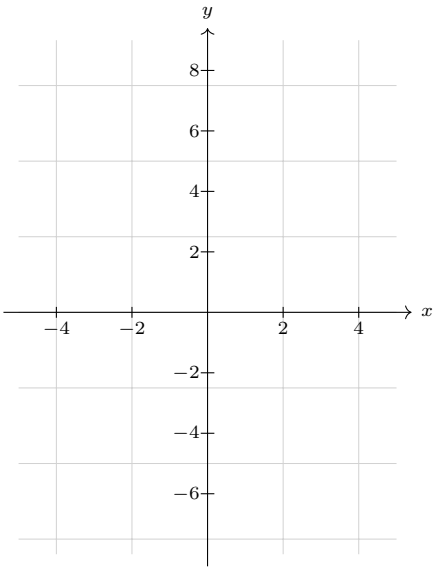
(e) At what other time during the dive does the plane reach this same height?

11. A relation is given by the table of values below.

x	-3	-1	0	2	4
y	8	4	2	-2	-6

(a) Write the relation as a set of ordered pairs.

(b) Plot the relation on the grid.
(c) Name the quadrant each plotted point lies in.
(One point lies on an axis — say which.)



(d) Should the points be joined? What would you need to know to decide?

(e) State the domain and the range.

(f) The points lie on a straight line. Write its equation in function notation.

(g) State the x-intercept and the y-intercept of that line as ordered pairs.

12. A ball is thrown upward from a 1 m platform. Its height is $h(t) = -5t^2 + 20t + 1$, where h is in metres and t is in seconds.

(a) Complete the table of values.

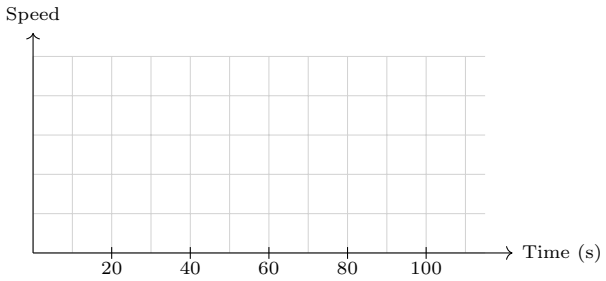
t (s)	0	1	2	3	4
h (m)					

(b) Use your table to find the greatest height reached and the time it occurs.

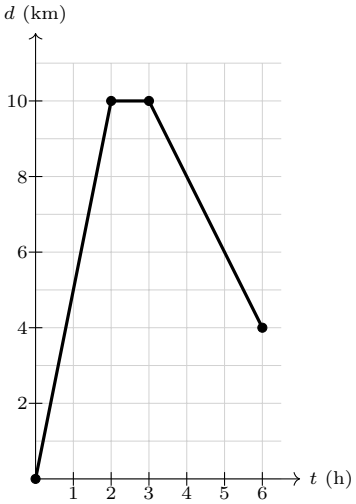
(c) The ball lands when $h = 0$. Evaluate $h(4.1)$, and use it with your table to explain why the ball lands between $t = 4$ s and $t = 4.1$ s.

(d) The ball lands at about $t = 4.05$ s. State the domain of the function in this context.

13. A cyclist starts from rest, speeds up steadily for 20 s, holds a steady speed for 40 s, slows steadily to a stop over the next 20 s, then stays stopped for 30 s. Sketch a graph of the cyclist's *speed* against time, and label each of the four stages.



14. The graph below shows Amira's distance from home during a walk.

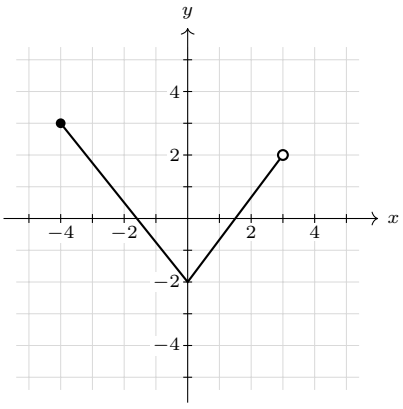


(a) Describe what Amira is doing in each of the three segments.

(b) Determine her rate of change, with units, for each segment.

(c) Explain what the sign of the rate of change in the third segment tells you.

15. State the domain and range of the relation graphed below. The left endpoint is solid and the right endpoint is hollow.



16. A phone's battery charge, in percent, after t hours of video streaming is modelled by the linear function $B(t) = 100 - 12t$.

(a) Evaluate $B(3)$ and explain its meaning.

(b) Solve $B(t) = 40$.

(c) After how many hours is the battery empty? (nearest hundredth)

(d) State a realistic domain and range.

Extension — beyond the Math 10C outcomes

17. *(Extension)* If $f(x) = 2^x$ and $f(-a) = \dfrac{1}{32}$, then $a =$

A. 5 B. -5 C. $-\dfrac{1}{5}$ D. $\dfrac{1}{5}$

18. *(Extension)* If $f(x) = 2 - 3x - 4x^2$, and if $f(x+3)$ is written in the form $ax^2 + bx + c$, the value of $a - b - c$ is _____.

Record your answer left to right. | | | | |

19. *(Extension)* $f(a) = \dfrac{a}{a+3}$. The exact value of $f(4) - f\left(\dfrac{1}{4}\right)$, written as a rational number in simplest form, is $\dfrac{p}{q}$. The value of p is _____.

Record your answer left to right. | | | | |

20. *(Extension)* To the nearest hundredth, determine the positive x-intercept of the graph of $3x^2 + 5y^2 = 9$.

Answer Key — Lesson Assignments

Lesson 1 Assignment

1. (a) x; y (b) ordered pairs (c) coordinates (d) origin (e) quadrants. **2.** (a) i) A ii) I iii) E (b) i) $C(4,2)$ ii) $G(-9,-8)$ iii) $F(-2,0)$ iv) $O(0,0)$ v) $H(1,-4)$ (c) i) C, D ii) A, B iii) G iv) H, I (d) E, F, O (e) E and O (both $x = 0$); they lie on the same vertical line (f) B, C and E (all $y = 2$), and F and O (both $y = 0$); points sharing a y-coordinate lie on the same horizontal line. **3.** A pentagon shaped like a simple house — a rectangle with a triangular roof. **4.** (a) S: $3, 5, 7, 9, 11$ (c) No — the data is discrete; intermediate values have no meaning (d) the number of dots is twice the figure number, plus one (e) $S = 2N + 1$ (f) 101 (g) 68. **5.** (b) Yes — the data is continuous (c) \$9.50 (d) 7 km (e) $F = 3.50 + 1.20d$ (f) when $d = 5$, $F =$ \$9.50; when $F = 11.90$, $d = 7$ km — matching the estimates (g) 3.50 is the fixed pickup fee (the fare when $d = 0$); 1.20 is the cost per kilometre. **6.** (a) P: $6, 10, 14, 18, 22$ (b) No — the data is discrete (c) $P = 4D + 2$ (d) 42 cm (e) diagram 19. **7.** Connect the points when the variables are continuous, i.e. every in-between value has meaning; leave them unconnected when the variables are discrete. **8.** C. **9.** B. **10.** B. **11.** B. **12.** 37.

Lesson 2 Assignment

1. (i) X (ii) Y (iii) 6, 10, 14 (iv) 25, 40. **2.** (i) p (ii) q (iii) 4, 9, 14 (iv) 11, 26, 41. **3.** (i) a (ii) b (iii) 3, 6, 9, 12 (iv) 14, 26, 38, 50. **4.** (a) independent h, dependent w (b) independent distance, dependent elevation (c) independent t, dependent n. **5.** (a) -10, -6, -4, 0, 6 (b) $x = 7$ (c) linear — x appears only to the first power, so equal steps in x give equal changes in y and the graph is a straight line. **6.** (a) 4, -4, -5, -1, 11 (b) 11 (c) $(-4)^2 - 5 = 11$, confirming (b) (d) nonlinear — the equation contains x^2, so its graph is a curve. **7.** (a) $+3$ each step: linear (b) $+1, +3, +5, +7$: nonlinear (c) -3 each step: linear. **8.** (a) $h = 20 - 1.5t$ (b) independent t, dependent h (c) 11 cm (d) 10 hours. **9.** (a) 55, 70, 85, 100 (b) $C = 40 + 15m$ (c) independent m, dependent C — the total cost depends on the number of months. **10.** B. **11.** A. **12.** 63. **13.** 43.

Lesson 3 Assignment

1. (a) -8 (b) -20 (c) 4 (d) 2 (e) -25 (f) 8.36. **2.** (a) 6 (b) 3 (c) 4 (d) -2 (e) ± 4 (f) 5. **3.** (a) x-int pt $= (-4, 0)$, y-int pt $= (0, 20)$ (b) x-int pt $= (3, 0)$, y-int pt $= (0, 24)$ (c) x-int pt $= (-3, 0)$, y-int pt $= (0, 6)$ (d) x-int pt $= (6, 0)$ and $(-6, 0)$, y-int pt $= (0, -12)$ (e) x-int pt $= (3, 0)$ and $(-3, 0)$, y-int pt $= (0, 9)$ (f) x-int pt $= (0, 0)$, y-int pt $= (0, 0)$ (g) no x-int, y-int pt $= (0, 9)$ (h) x-int pt $= (0, 0)$ and $(4, 0)$, y-int pt $= (0, 0)$ (i) x-int pt $= (2, 0)$ and $(-2, 0)$, y-int pt $= (0, 8)$. **4.** (a) $C = 150$, 350, 650, 950 (b) the flat fee charged before any distance is travelled (c) $n = -600$; not applicable since a distance cannot be negative (d) i) \$450 ii) \$1400 (e) 2600 km. **5.** (a) $V = 2400$, 1800, 1200, 600, 0 (b) 2400 L (c) 40 minutes (d) 900 L (e) 35 minutes (f) linear; V changes at the constant rate -60 L/min. **6.** (a) 80 m (b) $h = 35$, 60, 75, 80, 75, 60, 35, 0 (c) nonlinear (d) 7 seconds (e) the platform is 35 m above the ground at launch (f) the ball lands, 7 seconds after launch (g) 68.75 m (h) no to the left (time cannot be negative); no to the right (the ball has already landed). **7.** (a) any line through $(4, 0)$ and $(0, -2)$ (b) $y = -3$ (horizontal) (c) $x = 3$ (vertical) (d) $x = 0$ (the y-axis). **8.** B. **9.** 14. **10.** *(Extension)* (a) x-int pt $= (4, 0)$ and $(-4, 0)$, y-int pt $= (0, 8)$ and $(0, -8)$ (b) x-int pt $= (4, 0)$ and $(-4, 0)$, no y-int (c) x-int pt $= (7, 0)$ and $(-7, 0)$, y-int pt $= (0, 7)$ and $(0, -7)$. **11.** *(Extension)* 400 ($a = 4$, $b = -4$, $c = 5$, $d = -5$).

Lesson 4 Assignment

1. (a) $D = \{-1, 1, 3, 5, 7\}$, $R = \{-2, 4, 8\}$ (b) $D = \{-6\}$, $R = \{-3, 2, 5, 9\}$. **2.** $D = \{-3, -1, 0, 2, 4, 6\}$, $R = \{-8, -4, 0, 4, 6, 10\}$. **3.** $D = \{2, 4, 6, 8\}$, $R = \{10, 20\}$. **4.** (a) $D = \{-3, -1, 0, 2, 3\}$, $R = \{-2, -1, 2, 3, 4\}$ (b) $D = \{x \mid -5 \le x \le 3,\ x \in \mathbb{R}\}$, $R = \{y \mid -3 \le y \le 4,\ y \in \mathbb{R}\}$ (c) $D = \{x \mid x \in \mathbb{R}\}$, $R = \{y \mid y \ge -2,\ y \in \mathbb{R}\}$ (d) $D = \{x \mid -4 < x \le 3,\ x \in \mathbb{R}\}$, $R = \{y \mid -2 \le y < 2,\ y \in \mathbb{R}\}$ (e) $D = \{x \mid x \in \mathbb{R}\}$, $R = \{3\}$ (f) $D = \{-2\}$, $R = \{y \mid y \in \mathbb{R}\}$ (g) $D = \{x \mid x \in \mathbb{R}\}$, $R = \{y \mid y \in \mathbb{R}\}$ (h) $D = \{x \mid x > -2,\ x \in \mathbb{R}\}$, $R = \{y \mid y > -1,\ y \in \mathbb{R}\}$ (i) $D = \{x \mid -3 < x < 3,\ x \in \mathbb{R}\}$, $R = \{y \mid -1 \le y < 4,\ y \in \mathbb{R}\}$. **5.** (a) $D = \{x \mid x \in \mathbb{R}\}$, $R = \{y \mid y \le 3,\ y \in \mathbb{R}\}$ (b) $D = \{p \mid -5 \le p \le 5,\ p \in \mathbb{R}\}$, $R = \{q \mid -3 \le q \le 3,\ q \in \mathbb{R}\}$ (c) $D = \{-2, 0, 3\}$, $R = \{6\}$ (d) $D = \{x \mid -9 \le x \le 1,\ x \in \mathbb{R}\}$, $R = \{y \mid -4 \le y \le 6,\ y \in \mathbb{R}\}$. **6.** (a) finite list: $D = \{-2, 0, 2, 4\}$, $R = \{3\}$ (b) inequality: $D = \{x \mid -3 \le x \le 5,\ x \in \mathbb{R}\}$, $R = \{y \mid -7 \le y \le 9,\ y \in \mathbb{R}\}$ (c) inequality: $D = \{x \mid x \in \mathbb{R}\}$, $R = \{y \mid y \ge 0,\ y \in \mathbb{R}\}$. **7.** (a) $C = 14$, 26, 38, 53, 68 dollars (b) $D = \{d \mid 2 \le d \le 20,\ d \in \mathbb{R}\}$, $R = \{C \mid 14 \le C \le 68,\ C \in \mathbb{R}\}$; the company will not travel under 2 km or over 20 km, and cost rises with distance. **8.** B. **9.** 15 ($a = 5$, $b = 3$, $a \times b = 15$).

Lesson 5 Assignment

1. (a) function (x-values all distinct) (b) function (passes the vertical line test) (c) not a function ($5 \to 1$ and $5 \to 3$) (d) not a function ($x = 3$ gives 4 and 9). **2.** (a) function (b) not a function (c) not a function (d) not a function (e) function. **3.** (a) function (b) not a function (c) function (d) function (e) not a function (f) not a function. **4.** (a) not a function (Mrs. Y maps to three children) (b) function (each child has exactly one mother). **5.** C. **6.** A function is a relation in which no input has more than one output, so every function is a relation; a relation that pairs an input with two outputs is not a function (e.g. 3(a) is a function; 3(b) is not).

Lesson 6 Assignment

1. (a) $\{-3, 1, 5, 9\}$ (b) the four points $(0, -3)$, $(1, 1)$, $(2, 5)$, $(3, 9)$. **2.** (a) $\{-9, -2, -1, 0, 7\}$ (b) the five points $(-2, -9)$, $(-1, -2)$, $(0, -1)$, $(1, 0)$, $(2, 7)$. **3.** (a) $f(x) = 12, 7, 4, 4, 7, 12$ (b),(c) a parabola with vertex $(0, 3)$. **4.** (a) $(0, 9)$, $(5, 9)$, $(15, 9)$ (b) three points on the horizontal line $y = 9$. **5.** (a) name (b) inputs (c) outputs (d) outputs (e) independent (f) dependent (g) dependent. **6.** (a) 3 (b) -17 (c) -9. **7.** (a) -2 (b) -18 (c) 5. **8.** (a) 25 (b) -66. **9.** (a) 22 (b) 7. **10.** (a) 3 (b) 27 (c) 7.5 (d) $12 - 9t$ (e) $18 - 3a$. **11.** (a) $x = 7$ (b) $x = -4$ (c) $t = 7$ (d) $x = -6$ (e) $x = \pm 5$. **12.** (a) -20 (b) $27x - 29$ (c) $x = 7$. **13.** (a)(i) 5 (ii) $\frac{1}{7}$ (iii) 4 (b) $x = 121$. **14.** 43. **15.** (a) \$17.90 (b) a 12 km trip costs \$17.90 (c) $d = 10$ km (d) the fixed pickup fee. **16.** D. **17.** B. **18.** 18. **19.** *(Extension)* $a^3 - 2$. **20.** *(Extension)* $a = \pm 4\sqrt{3}$. **21.** *(Extension)* (a) $2x^2 + 5x - 3$ (b) $2x^2 + 11x + 9$.

Lesson 7 Assignment

1. (a) $y = 12 - 5x$ (b) $y = 3x^2 + 7$ (c) $y = 4t - 15$. **2.** (a) $f(x) = 6x + 13$ (b) $f(w) = 8w - 21$ (c) $f(x) = 3 - \frac{3}{4}x$. **3.** (a) $f(x) = 0.25x + 1.75$ (b) $f(t) = 55t$. **4.** (a) 0, 2, −2 (b) $(-6, 0)$, $(-1, 2)$, $(5, -2)$ (c) i) $x = -8, 0, 8$ ii) $x = -4, 4$ iii) no solution (d) $q(-8) = 4$, $q(8) = 4$ (e) x-int: $q(-6) = q(-2) = q(2) = q(6) = 0$; y-int: $q(0) = 4$ (f) $D = \{x \mid -8 \le x \le 8,\ x \in \mathbb{R}\}$, $R = \{q(x) \mid -4 \le q(x) \le 4,\ q(x) \in \mathbb{R}\}$. **5.** (a) $R = \{-7, -5, 1\}$ (b) $x = -2$ or $x = 2$. **6.** (a) $(9, 12)$, $(3, 0)$, $(-6, -18)$, $(6, 6)$, $(-3, -12)$, $(0, -6)$ (b) $x = 8$; the line keeps rising, so each output height is reached by exactly one input. **7.** (a) 23 (b) 22 (c) x-int $\frac{11}{3}$, y-int 11. **8.** (a) i) the function is named s, not t, and the point was counted in grid boxes instead of the 5-unit scale — it is $(-15, 20)$ ii) $s(-15) = 20$ (b) i) $x = -30, 10$ ii) $x = -10, 30$ (c) i) −10 ii) 30 (d) $D = \{x \mid -30 \le x \le 30,\ x \in \mathbb{R}\}$, $R = \{s(x) \mid -10 \le s(x) \le 30,\ s(x) \in \mathbb{R}\}$ (e) exactly two solutions only at the extreme heights: $b = 30$ with $a = -10$ or 30; $b = -10$ with $a = -30$ or 10. **9.** *(Extension)* $a = 625$. **10.** B. **11.** C. **12.** 2.5. **13.** (b) 19 cm (c) 8 days after treatment begins the seedling is 19 cm tall (d) i) 5 ii) 33 iii) 47 (e) 12 days (f) $D = \{t \mid 0 \le t \le 24,\ t \in \mathbb{R}\}$, $R = \{H(t) \mid 5 \le H(t) \le 47,\ H(t) \in \mathbb{R}\}$. **14.** (b) \$2555 (c) printing 60 t-shirts costs \$2555 (d) $C(0) = 2150$ — the fixed setup cost paid before any shirt is printed (e) 600 (f) \$1675. **15.** (b) 260 (c) when the average daily temperature is −10 °C, 260 packets are sold (d) 5 °C (e) the lower limit is the coldest average daily temperature the resort realistically experiences — take it from climate records, not from the equation (f) answers vary: $D = \{t \mid -20 \le t \le 22.5,\ t \in \mathbb{R}\}$, $R = \{P(t) \mid 0 \le P(t) \le 340,\ P(t) \in \mathbb{R}\}$. **16.** (b) 857.5 m (c) the case lands at $t = 20$ s, so $h(25) = -1102.5$ is negative and meaningless (d) 1960 m (e) 20 s (f) $D = \{t \mid 0 \le t \le 20,\ t \in \mathbb{R}\}$, $R = \{h(t) \mid 0 \le h(t) \le 1960,\ h(t) \in \mathbb{R}\}$.

Lesson 8 Assignment

1. (a) Line 2: $\frac{1}{4}$ hr, 24 km/hr, Derek; Line 3: $\frac{3}{2}$ hr, 4 km/hr, Grace; Line 1 is Farah. (b) W: all three leave the mall together; X: Derek (cyclist) reaches the stadium and stops; Y: Farah (jogger) reaches the stadium and stops; Z: Grace (walker) reaches the stadium. (c) Steeper line ⇒ greater speed: steepest is the cyclist, then the jogger, then the walker. **2.** The ball rises to a maximum height of about 8 m at about 25 m out, first lands at 54 m, bounces twice with decreasing height, then rolls to a stop at about 60 m. **3.** Left at 8:00 with $\frac{3}{4}$ tank; drove about an hour; stopped about 45 min; drove until 11:00 ($\frac{1}{4}$ tank); longer break 11:00–12:30; drove to 13:00 on a nearly empty tank; refuelled to full; parked until 16:00; drove home, arriving about 18:00 with about half a tank. **4.** The Comet is older than the Aurora and has the lower resale value; the Comet tows more, but the Aurora is more fuel efficient. **5.** A first fall from 12 m, then arches of height 7.8, 5.07, 3.30 and 2.14 m, each touching $h = 0$. **6.** (a) Ocean tide height over a day (high tide about every 12 hours). (b) A skydiver's speed after the parachute opens — it drops quickly, then levels off at a steady landing speed. **7.** Sketches vary: (a) decreasing, flattening; (b) a decreasing staircase reaching \$0 after 8 months; (c) rises from a cold morning to an early-afternoon peak, then falls; (d) a repeating wave, three cycles, never reaching zero. **8.** The path doubles back, so a vertical line meets it three times — the student would be three distances from her locker at one instant. It fails the vertical line test. **9.** A. **10.** E **11.** C **12.** F **13.** A **14.** (a) a smooth decreasing curve through the seven points (b) yes — time and temperature are continuous (c) $D = \{t \mid 0 \le t \le 30,\ t \in \mathbb{R}\}$, $R = \{T \mid 27 \le T \le 80,\ T \in \mathbb{R}\}$; the temperature levels off at room temperature, so the curve cannot be extended indefinitely. **15.** B. **16.** 75.

Lesson 9 Assignment

1. C. **2.** A. **3.** C. **4.** B. **5.** A ($p(-2) = 13,\ 10,\ 8,\ -3$ for functions $4, 3, 2, 1$). **6.** B. **7.** B. **8.** 8. **9.** 16. **10.** table: 84, 77, 72, 69, 68, 69, 72, 77, 84 (a) 84 m (b) 68 m (c) $t = 4$ s (d) 72 m (e) $t = 6$ s. **11.** (a) $\{(-3, 8), (-1, 4), (0, 2), (2, -2), (4, -6)\}$ (b) five separate points (c) $(-3, 8)$ and $(-1, 4)$ in QII; $(2, -2)$ and $(4, -6)$ in QIV; $(0, 2)$ lies on the y-axis (d) not from the table alone — you need to know what x and y represent (join only if x is continuous) (e) $D = \{-3, -1, 0, 2, 4\}$, $R = \{-6, -2, 2, 4, 8\}$ (f) $f(x) = -2x + 2$ (g) x-int $(1, 0)$, y-int $(0, 2)$. **12.** (a) 1, 16, 21, 16, 1 (b) 21 m at $t = 2$ s (c) $h(4.1) = -1.05 < 0$ while $h(4) = 1 > 0$, so $h = 0$ between 4 and 4.1 s (d) $D = \{t \mid 0 \le t \le 4.05,\ t \in \mathbb{R}\}$. **13.** Rising straight line 0–20 s (speeding up), horizontal 20–60 s (steady speed), falling straight line to the axis 60–80 s (slowing), horizontal along the axis 80–110 s (stopped). **14.** (a) walks away from home at a steady rate; stops for an hour; returns toward home at a steady, slower rate (b) 5 km/h, 0 km/h, −2 km/h (c) the negative sign means her distance from home is decreasing — she is walking back toward home. **15.** $D = \{x \mid -4 \le x < 3,\ x \in \mathbb{R}\}$, $R = \{y \mid -2 \le y \le 3,\ y \in \mathbb{R}\}$. **16.** (a) 64: after 3 h the battery is at 64% (b) $t = 5$ h (c) 8.33 h (d) $D = \{t \mid 0 \le t \le 8.33,\ t \in \mathbb{R}\}$, $R = \{B \mid 0 \le B \le 100,\ B \in \mathbb{R}\}$. **17.** *(Extension)* A. **18.** *(Extension)* 66 ($f(x+3) = -4x^2 - 27x - 43$, so $a - b - c = -4 + 27 + 43 = 66$). **19.** *(Extension)* 45 ($\frac{4}{7} - \frac{1}{13} = \frac{45}{91}$). **20.** *(Extension)* 1.73.